An Analysis of Uncertainty quantification

Uncertainty quantification

Uncertainty quantification -- Part 1

R δ (x , x ′) = exp ■ { − ∑ k = 1 d ω k δ (x k − x k ′) 2 } . {\displaystyle R^{\delta }(\mathbf {x} ,\mathbf {x} ')=\exp \left\{-\sum _{k=1}^{d}\omega _{k}^{\delta }(x_{k}-x_{k}')^{2}\right\}.}

Uncertainty quantification -- Part 2

Selective methodologies Much research has been done to solve uncertainty quantification problems, though a majority of them deal with uncertainty propagation. During the past one to two decades, a number of approaches for inverse uncertainty quantification problems have also been developed and have proved to be useful for most small- to medium-scale problems.

Uncertainty quantification -- Part 3

Dimensionality issue: The computational cost increases dramatically with the dimensionality of the problem, i.e. the number of input variables and/or the number of unknown parameters. Identifiability issue: Multiple combinations of unknown parameters and discrepancy function can yield the same experimental prediction. Hence different values of parameters cannot be distinguished/identified. This issue is circumvented in a Bayesian approach, where such combinations are averaged over. Incomplete model response: Refers to a model not having a solution for some combinations of the input variables. Quantifying uncertainty in the input quantities: Crucial events missing in the available data or critical quantities unidentified to analysts due to, e.g., limitations in existing models. Little consideration of the impact of choices made by analysts.

Uncertainty quantification -- Part 4

Together with the prior distribution of unknown parameters, and data from both computer models and experiments, one can derive the maximum likelihood estimates for { β δ , σ δ , ω k δ , k = 1 , … , d } {\displaystyle \left\{{\boldsymbol {\beta }}^{\delta },\sigma _{\delta },\omega _{k}^{\delta },k=1,\ldots ,d\right\}} . At the same time, β m {\displaystyle {\boldsymbol {\beta }}^{m}} from Module 1 gets updated as well.

Uncertainty quantification -- Part 5

To evaluate low-order moments of the outputs, i.e. mean and variance. To evaluate the reliability of the outputs. This is especially useful in reliability engineering where outputs of a system are usually closely related to the performance of the system. To assess the complete probability distribution of the outputs. This is useful in the scenario of utility optimization where the complete distribution is used to calculate the utility. To estimate uncertainty in values, that cannot be directly measured, when calculated with experimentally measurable values. This is especially useful in experimental physics and chemistry contexts.

Uncertainty quantification -- Part 6

Mathematical perspective In mathematics, uncertainty is often characterized in terms of a probability distribution. From that perspective, aleatoric uncertainty means that the probability distribution does not unambiguously point to a specific outcome or not being able to predict what a random sample drawn from a probability distribution will be, while epistemic uncertainty means that one does not know the relevant probability distribution because one lacks a basis for determining it. The figure below shows four different examples of aleatoric and epistemic uncertainty, visualized in two mathematically equivalent ways consisting of a triangle of subjective opinions from subjective logic and as a Beta probability density function.

Uncertainty quantification -- Part 7

where y m (x , θ) {\displaystyle y^{m}(\mathbf {x} ,{\boldsymbol {\theta }})} denotes the computer model response that depends on several unknown model parameters θ {\displaystyle {\boldsymbol {\theta }}} , and θ ∗ {\displaystyle {\boldsymbol {\theta }}^{*}} denotes the true values of the unknown parameters in the course of experiments. The objective is to either estimate θ ∗ {\displaystyle {\boldsymbol {\theta }}^{*}} , or to come up with a probability distribution of θ ∗ {\displaystyle {\boldsymbol {\theta }}^{*}} that encompasses the best knowledge of the true parameter values.

Uncertainty quantification -- Part 8

R m ((x , θ) , (x ′ , θ ′)) = exp ■ { − ∑ k = 1 d ω k m (x k − x k ′) 2 } exp ■ { − ∑ k = 1 r ω d + k m (θ k − θ k ′) 2 } . {\displaystyle R^{m}{\big (}(\mathbf {x} ,{\boldsymbol {\theta }}),(\mathbf {x} ',{\boldsymbol {\theta }}'){\big)}=\exp \left\{-\sum _{k=1}^{d}\omega _{k}^{m}(x_{k}-x_{k}')^{2}\right\}\exp \left\{-\sum _{k=1}^{r}\omega _{d+k}^{m}(\theta _{k}-\theta _{k}')^{2}\right\}.}

Uncertainty quantification -- Part 9

For most cases, p (θ , φ ■ data) {\displaystyle p({\boldsymbol {\theta }},{\boldsymbol {\varphi }}\mid {\text{data}})} is a highly intractable function of φ {\displaystyle {\boldsymbol {\varphi }}} . Hence the integration becomes very troublesome. Moreover, if priors for the other hyperparameters φ {\displaystyle

{\boldsymbol {\varphi }}} are not carefully chosen, the complexity in numerical integration increases even more. In the prediction stage, the prediction (which should at least include the expected value of system responses) also requires numerical integration. Markov chain Monte Carlo (MCMC) is often used for integration; however it is computationally expensive. The fully Bayesian approach requires a huge amount of calculations and may not yet be practical for dealing with the most complicated modelling situations.

Uncertainty quantification -- Part 10

Types of problems There are two major types of problems in uncertainty quantification: one is the forward propagation of uncertainty (where the various sources of uncertainty are propagated through the model to predict the overall uncertainty in the system response) and the other is the inverse assessment of model uncertainty and parameter uncertainty (where the model parameters are calibrated simultaneously using test data). There has been a proliferation of research on the former problem and a majority of uncertainty analysis techniques were developed for it. On the other hand, the latter problem is drawing increasing attention in the engineering design community, since uncertainty quantification of a model and the subsequent predictions of the true system response(s) are of great interest in designing robust systems.

Uncertainty quantification -- Part 11

where y e (x) {\displaystyle y^{e}(\mathbf {x})} denotes the experimental measurements as a function of several input variables x {\displaystyle \mathbf {x} } , y m (x) {\displaystyle y^{m}(\mathbf {x})} denotes the computer model (mathematical model) response, δ (x) {\displaystyle \delta (\mathbf {x})} denotes the additive discrepancy function (aka bias function), and ε {\displaystyle \varepsilon } denotes the experimental uncertainty. The objective is to estimate the discrepancy function δ (x) {\displaystyle \delta (\mathbf {x})} , and as a by-product, the resulting updated model is y m (x) + δ (x) {\displaystyle y^{m}(\mathbf {x})+\delta (\mathbf {x})} . A prediction confidence interval is provided with the updated model as the quantification of the uncertainty.